

Non-commutative deformations of quotient singularities.

Lien Cartaya Bermúdez

Stephen Griffeth, PhD.
Advisor

Thesis for Math Master's Degreee
04.03.2024



We thanks to the Agencia Nacional de Investigación y Desarrollo, ANID, Chile, for supporting this work through the FONDECYT Proyecto Regular 1231355.

Review of the key results in *Non-commutative deformations of Kleinian singularities*, by Crawley-Boevey and Holland [[CH98](#)] (1998).

Table of Contents

- 1 Context.
- 2 Main results.
- 3 An example.
- 4 An application of the results.
- 5 Principal bibliography.

Deformation of the Kleinian singularity

K a field, $\overline{K} = K$, $\text{char} K = 0$.

$\Gamma \leq \text{SL}_2(K)$ a non-trivial finite subgroup.

Deformation of the Kleinian singularity

K a field, $\overline{K} = K$, $\text{char} K = 0$.

$\Gamma \leq \text{SL}_2(K)$ a non-trivial finite subgroup.

$K^2 \twoheadrightarrow K^2/\Gamma = \text{Spec}(K[x, y]^\Gamma)$ the Kleinian singularity.

We study non-commutative deformations of $K[x, y]^\Gamma$, called $\mathcal{O}^\lambda$.

$$\mathcal{O}^\lambda := e \frac{K\langle x, y \rangle \rtimes \Gamma}{(xy - yx - \lambda)} e, \quad e = \frac{1}{|\Gamma|} \sum_{\gamma \in \Gamma} \gamma, \quad \lambda \in Z(K\Gamma).$$

McKay graph

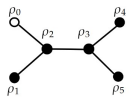
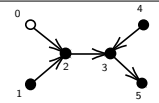
Group	McKay graph	Type
C_n		A_{n-1}^1
D_n^*		D_{n+2}^1
T^*		E_6^1
O^*		E_7^1
I^*		E_8^1

Irreducible representations, N_i , $i \in I$ of $\Gamma \iff$ Vertices set I

Multiplicity of N_i in $V \otimes N_j \iff$ edge between i and j

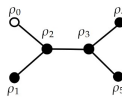
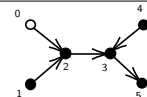
Extended Dynkin quiver

Q an extended Dynkin quiver associated to Γ .

$\Gamma \leq \mathrm{SL}_2(K)$	$D_3 = \langle a, b a^3 = b^2 = (ba)^2 = 1 \rangle$
Dynkin extended graph	
Quiver Q	
$\delta = (\delta)_{i \in I} = (\dim N_i)_{i \in I}$	$\delta = (1, 1, 2, 2, 1, 1)$

Extended Dynkin quiver

Q an extended Dynkin quiver associated to Γ .

$\Gamma \leq \mathrm{SL}_2(K)$	$D_3 = \langle a, b a^3 = b^2 = (ba)^2 = 1 \rangle$
Dynkin extended graph	
Quiver Q	
$\delta = (\delta)_{i \in I} = (\dim N_i)_{i \in I}$	$\delta = (1, 1, 2, 2, 1, 1)$

$$Z(K\Gamma) = K', \lambda \leftrightarrow (\lambda_i), \lambda_i \text{ is the trace of } \lambda \text{ on } N_i.$$

Deformed preprojective algebra

For an arbitrary quiver Q , with vertices set I and arrows set Q ,

$\Pi(Q) := K\overline{Q}/(\sum_{a \in Q} [a, a^*])$ the preprojective algebra of Q , $K\overline{Q}$ the path algebra.

Deformed preprojective algebra

For an arbitrary quiver Q , with vertices set I and arrows set Q ,

$\Pi(Q) := K\overline{Q}/(\sum_{a \in Q} [a, a^*])$ the preprojective algebra of Q , $K\overline{Q}$ the path algebra.

We study the deformed preprojective algebra $\Pi^\lambda(Q) := K\overline{Q}/(\sum_{a \in Q} [a, a^*] - \sum_{i \in I} \lambda_i e_i)$,

$\lambda = (\lambda_i)_{i \in I} \in K^I$, e_i the trivial path at vertex i .

Table of Contents

- 1 Context.
- 2 Main results.**
- 3 An example.
- 4 An application of the results.
- 5 Principal bibliography.

Theorem PBW

$K\langle x, y \rangle \rtimes \Gamma := K\langle x, y \rangle \otimes_K K\Gamma$ the semi-direct product algebra.

$\mathcal{S}^\lambda := (K\langle x, y \rangle \rtimes \Gamma) / (xy - yx - \lambda), \lambda \in Z(K\Gamma)$

$\mathcal{O}^\lambda := e\mathcal{S}^\lambda e, e = \frac{1}{|\Gamma|} \sum_{\gamma \in \Gamma} \gamma.$

Theorem PBW

$K\langle x, y \rangle \rtimes \Gamma := K\langle x, y \rangle \otimes_K K\Gamma$ the semi-direct product algebra.

$\mathcal{S}^\lambda := (K\langle x, y \rangle \rtimes \Gamma) / (xy - yx - \lambda), \lambda \in Z(K\Gamma)$

$\mathcal{O}^\lambda := e\mathcal{S}^\lambda e$, $e = \frac{1}{|\Gamma|} \sum_{\gamma \in \Gamma} \gamma$.

$\mathcal{S}^\lambda$ has a filtration, x and y have degree 1 and $\gamma \in \Gamma$ has degree 0.

Theorem

$gr\mathcal{S}^\lambda \cong K[x, y] \rtimes \Gamma$ and $gr\mathcal{O}^\lambda \cong K[x, y]^\Gamma$.

Morita equivalence

Q a Dynkin extended quiver associated to Γ .

$f_i \in K\Gamma$ an idempotent; $K\Gamma f_i \cong N_i$. The f_i are orthogonal.

$$f = f_0 + \dots + f_n, f_0 = e$$

Morita equivalence

Q a Dynkin extended quiver associated to Γ .

$f_i \in K\Gamma$ an idempotent; $K\Gamma f_i \cong N_i$. The f_i are orthogonal.

$$f = f_0 + \dots + f_n, f_0 = e$$

Theorem 3.8: $\Pi^\lambda(Q) \cong fS^\lambda f$.

Corollary 3.9

$\mathcal{O}^\lambda \cong e_0 \Pi^\lambda(Q) e_0$, e_0 the trivial path at vertex 0, and $S^\lambda \stackrel{M}{\sim} \Pi^\lambda(Q)$, $\forall \lambda \in Z(K\Gamma) = K^I$.

Ringel form and Weyl group for a quiver

$$\langle \alpha, \beta \rangle = \sum_{i \in I} \alpha_i \beta_i - \sum_{a \in Q} \alpha_{ta} \beta_{ha} \text{ the Ringel form, with } \alpha, \beta \in K^I.$$

$$(\alpha, \beta) = \langle \alpha, \beta \rangle + \langle \beta, \alpha \rangle \text{ the symmetrization of the Ringel form.}$$

Ringel form and Weyl group for a quiver

$$\langle \alpha, \beta \rangle = \sum_{i \in I} \alpha_i \beta_i - \sum_{a \in Q} \alpha_{ta} \beta_{ha} \text{ the Ringel form, with } \alpha, \beta \in K^I.$$

$$(\alpha, \beta) = \langle \alpha, \beta \rangle + \langle \beta, \alpha \rangle \text{ the symmetrization of the Ringel form.}$$

The *Weyl group* is generated by the *simple reflections* $s_i \in \text{Aut} \mathbb{Z}^I$, $\alpha \mapsto \alpha - (\alpha, \epsilon_i) \epsilon_i$.

Roots for a quiver

The *simple roots* are the coordinate vectors $\epsilon_i \in \mathbb{Z}^I$ for $i \in I$ a loopfree vertex.

- $\alpha \in \mathbb{Z}^I$ is a *real root* if $w \in W$ and a simple root ϵ_j with $\alpha = w\epsilon_j$.

Roots for a quiver

The *simple roots* are the coordinate vectors $\epsilon_i \in \mathbb{Z}^I$ for $i \in I$ a loopfree vertex.

- $\alpha \in \mathbb{Z}^I$ is a *real root* if $w \in W$ and a simple root ϵ_j with $\alpha = w\epsilon_j$.

$$F = \{\alpha \in \mathbb{N}^I \mid \alpha \neq 0, \text{supp}\alpha \text{ is connected}, (\alpha, \epsilon_i) \leq 0 \forall i\}$$

- $\alpha \in \mathbb{Z}^I$ is an *imaginary root* if $\exists w \in W$ and $\beta \in F$ with $\alpha = w\beta$ or $\alpha = -w\beta$.

A *root* is either real or imaginary.

Classification of simple $\Pi^\lambda(Q)$ -modules

Q an extended Dynkin quiver associated to Γ .

$\lambda \cdot \delta \neq 0$. $R_\lambda = \{\alpha \text{ a root} \mid \lambda \cdot \alpha = 0\}$.

Σ_λ the set of positive minimal elements of R_λ .

Classification of simple $\Pi^\lambda(Q)$ –modules

Q an extended Dynkin quiver associated to Γ .

$\lambda \cdot \delta \neq 0$. $R_\lambda = \{\alpha \text{ a root} \mid \lambda \cdot \alpha = 0\}$.

Σ_λ the set of positive minimal elements of R_λ .

Theorem 7.7

$$\begin{array}{ccc} \{\text{Isoclasses of finite-dimensional simple } \Pi^\lambda(Q)\text{--modules}\} & \xrightarrow{\sim} & \Sigma_\lambda \\ M & \mapsto & \dim M \end{array}$$

Table of Contents

- 1 Context.
- 2 Main results.
- 3 An example.**
- 4 An application of the results.
- 5 Principal bibliography.

Cyclic group of order 2

$$\Gamma = C_2 = \langle \sigma \mid \sigma^2 = 1 \rangle.$$

Letting $K = \mathbb{C}$, $\lambda = \sigma \in Z(\mathbb{C}C_2)$:

$$\mathcal{S}^\lambda = \frac{\mathbb{C}\langle x, y \rangle \rtimes C_2}{(xy - yx - \sigma)}, \quad \mathcal{O}^\lambda = \frac{1 + \sigma}{2} \mathcal{S}^\lambda \frac{1 + \sigma}{2}$$

Cyclic group of order 2

$$\Gamma = C_2 = \langle \sigma \mid \sigma^2 = 1 \rangle.$$

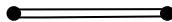
Letting $K = \mathbb{C}$, $\lambda = \sigma \in Z(\mathbb{C}C_2)$:

$$\mathcal{S}^\lambda = \frac{\mathbb{C}\langle x, y \rangle \rtimes C_2}{(xy - yx - \sigma)}, \quad \mathcal{O}^\lambda = \frac{1 + \sigma}{2} \mathcal{S}^\lambda \frac{1 + \sigma}{2}$$

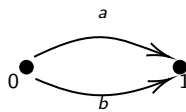
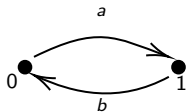
C_2	1	σ
χ_0	$1 = \dim N_0$	1
χ_1	$1 = \dim N_1$	-1

Cyclic group of order 2

$$C_2 \leftrightarrow A_1^1$$



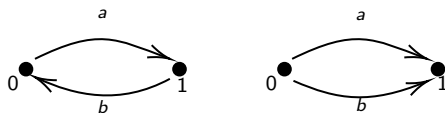
Two possible orientation for Q :



Cyclic group of order 2

$$C_2 \leftrightarrow A_1^1 \quad \bullet \text{---}\text{---}\text{---}\bullet$$

Two possible orientation for Q :



Q the Kronecker quiver (to the right), letting $\lambda \in \mathbb{C}^2$

$$\Pi^\lambda(Q) = \mathbb{C}\overline{Q}/(-a^*a - b^*b - \lambda_0 e_0, aa^* + bb^* - \lambda_1 e_1)$$

Kronecker quiver

$$\alpha = \delta + \phi = (2, 1) \in \mathbb{Z}^2.$$

$\delta = (1, 1)$ the fundamental root for Q , $\phi = (1, 0)$ a simple root for Q .

Kronecker quiver

$$\alpha = \delta + \phi = (2, 1) \in \mathbb{Z}^2.$$

$\delta = (1, 1)$ the fundamental root for Q , $\phi = (1, 0)$ a simple root for Q .

Take $\lambda = (1, -2)$, $\lambda \cdot \alpha = 0 \Rightarrow \lambda_1 = -2\lambda_0$,

$$\Pi^\lambda(Q) = \mathbb{C}\overline{Q}/(-a^*a - b^*b - e_0, aa^* + bb^* + 2e_1)$$

Kronecker quiver

$$M \in \text{Rep}(\overline{Q}), M(0) = \mathbb{C}^2, M(1) = \mathbb{C} \Rightarrow \dim M = \alpha = (2, 1)$$

$$v \in \mathbb{C}^2, w \in \mathbb{C},$$

$$M(a) : M(0) \rightarrow M(1), (v_0, v_1) \mapsto v_0,$$

$$M(b) : M(0) \rightarrow M(1), (v_0, v_1) \mapsto v_1,$$

$$M(a^*) : M(1) \rightarrow M(0), w \mapsto (-w, 0),$$

$$M(b^*) : M(1) \rightarrow M(0), w \mapsto (0, -w)$$

Kronecker quiver

$$M \in \text{Rep}(\overline{Q}), M(0) = \mathbb{C}^2, M(1) = \mathbb{C} \Rightarrow \dim M = \alpha = (2, 1)$$

$$v \in \mathbb{C}^2, w \in \mathbb{C},$$

$$M(a) : M(0) \rightarrow M(1), (v_0, v_1) \mapsto v_0,$$

$$M(b) : M(0) \rightarrow M(1), (v_0, v_1) \mapsto v_1,$$

$$M(a^*) : M(1) \rightarrow M(0), w \mapsto (-w, 0),$$

$$M(b^*) : M(1) \rightarrow M(0), w \mapsto (0, -w)$$

$$\begin{aligned} (-a^*a - b^*b - e_0) \cdot (v + w) &= 0, (aa^* + bb^* + 2e_1) \cdot (v + w) = 0 \\ \Rightarrow M &\in \text{Rep}(\Pi^\lambda(Q)) \end{aligned}$$

Table of Contents

- 1 Context.
- 2 Main results.
- 3 An example.
- 4 An application of the results.**
- 5 Principal bibliography.

Zero-fiber ring

F a field, V a finite-dimensional F -vector space.

$G \leq \mathrm{GL}(V)$ a finite group.

$F[V]^G$ the ring of functions on the scheme V/G .

$$F[V]^G \hookrightarrow F[V] \quad \longleftrightarrow \quad \pi : V \twoheadrightarrow V/G$$

Zero-fiber ring

$$\pi^{-1}(0) \subseteq V \iff F[\pi^{-1}(0)] = F[V]/\langle f - f(0) \mid f \in F[V]^G \rangle$$

zero-fiber

zero-fiber ring

Zero-fiber ring

$$\pi^{-1}(0) \subseteq V \quad \longleftrightarrow \quad F[\pi^{-1}(0)] = F[V]/\langle f - f(0) \mid f \in F[V]^G \rangle$$

zero-fiber

zero-fiber ring

$d = d(G)$ the *degree* of the zero-fiber is the dimension of the zero-fiber ring.

Zero-fiber ring of quaternionic quotient singularities

$$W_n(\Gamma) = \{\gamma_1^{(1)} \gamma_2^{(2)} \cdots \gamma_n^{(n)} \omega \mid \gamma_1, \gamma_2, \dots, \gamma_n \in \Gamma, \omega \in S_n\}$$

$$\gamma^{(p)} = \begin{pmatrix} 1 & 0 & & & 0 \\ 0 & 1 & & & 0 \\ & & \ddots & & \\ & & & 1 & \\ & \vdots & & \gamma & \vdots \\ & \vdots & & 1 & \vdots \\ 0 & & & & \ddots & 1 \end{pmatrix}, \text{ with } \gamma \in \Gamma \leq \mathbb{H}^\times \text{ in position } (p, p)$$

Seeing $S_n \leq \mathrm{GL}_n(\mathbb{H})$ the subgroup of permutation matrices.

Zero-fiber ring of quaternionic quotient singularities

Letting $V = \mathbb{H}^n$ and $G = W_n(\Gamma)$,

Theorem Zero-fiber of $\mathbb{H}^n \twoheadrightarrow \mathbb{H}^n/W_n(\Gamma)$ is of degree $d \geq ((n+1)|\Gamma| - 1)^n$.

Zero-fiber ring of quaternionic quotient singularities

Letting $V = \mathbb{H}^n$ and $G = W_n(\Gamma)$,

Theorem Zero-fiber of $\mathbb{H}^n \twoheadrightarrow \mathbb{H}^n/W_n(\Gamma)$ is of degree $d \geq ((n+1)|\Gamma| - 1)^n$.

By using Theorem 3.8, its Corollary 3.9 and Theorem 7.7.

$n = 1$ By a direct calculation. (=)

Zero-fiber ring of quaternionic quotient singularities

Letting $V = \mathbb{H}^n$ and $G = W_n(\Gamma)$,

Theorem Zero-fiber of $\mathbb{H}^n \twoheadrightarrow \mathbb{H}^n/W_n(\Gamma)$ is of degree $d \geq ((n+1)|\Gamma| - 1)^n$.

By using Theorem 3.8, its Corollary 3.9 and Theorem 7.7.

$n = 1$ By a direct calculation. (=)

$n > 1$ By using the PBW Theorem:

$$\begin{aligned} S^\lambda &\text{ is a deformation of } \mathbb{C}[x, y] \rtimes \Gamma \\ \Rightarrow S^{\lambda^{\otimes n}} \rtimes S_n &\text{ is a deformation of } \mathbb{C}[x_1, \dots, x_n, y_1, \dots, y_n] \rtimes W_n(\Gamma) \end{aligned}$$

Relate $\mathcal{S}^\lambda$ to $\Pi^\lambda(Q)$
 Q extended Dynkin

$\Rightarrow$
Quiver representation

Information about $\mathcal{S}^\lambda$

The article by Crawley-Boevey and Holland provides us with a tool to find representations of the deformation of the quaternionic quotient singularity (see [\[GC24\]](#))

Table of Contents

- 1 Context.
- 2 Main results.
- 3 An example.
- 4 An application of the results.
- 5 Principal bibliography.**

Principal bibliography

- [CH98] William Crawley-Boevey and Martin P. Holland. “Noncommutative deformations of Kleinian singularities”. In: *Duke Mathematical Journal* 92.3 (Apr. 1998). ISSN: 0012-7094. DOI: [10.1215/S0012-7094-98-09218-3](https://doi.org/10.1215/S0012-7094-98-09218-3). URL: <http://dx.doi.org/10.1215/S0012-7094-98-09218-3>.
- [GC24] Stephen Griffeth and Lien Cartaya. “Zero fibers of quaternionic quotient singularities”. In: (Jan. 2024). DOI: [10.48550/ARXIV.2402.00158](https://doi.org/10.48550/ARXIV.2402.00158). URL: <https://arxiv.org/abs/2402.00158>.

Q&A

A photograph of a classroom where several students are sleeping at their desks. The students are in the foreground and middle ground, with their heads resting on their arms or desks. The background shows a typical classroom setting with a whiteboard, a clock, and a bookshelf.

Thank you for your attention !